TITANIC DATASET

Exploratory Data Analysis (EDA) Report

Data overview • Missing values • Survival patterns • Correlations

Dataset size	891 rows × 12 columns
Passengers survived	342 (38.38%)
Passengers did not survive	549 (61.62%)
Project objective	Explore passenger data and identify patterns associated with survival.

Prepared from the analysis results supplied for this project. The report summarizes the reported findings; original notebook charts are not embedded.

1. Introduction

Exploratory Data Analysis (EDA) helps understand a dataset before drawing conclusions or building predictive models. This project examines Titanic passenger information, checks data quality, summarizes variables, and explores how survival varied by passenger class, gender, age, and fare.

Objectives

- Inspect dataset structure and size.
- Identify missing values and note reported handling.
- Study age and survival distributions.
- Compare survival across passenger classes and genders.
- Review numerical relationships and potential outliers.

2. Dataset Overview

Metric	Reported result
Rows / passengers	891
Columns	12
Age range	0.42 to 80 years
Mean age	Approximately 29.36 years
Median age	28 years
Non-survivors	549 (61.62%)
Survivors	342 (38.38%)

The dataset contains numerical fields such as Age and Fare and categorical fields such as Sex and Embarked. Survived is binary: 1 indicates survival and 0 indicates non-survival.

3. Data Quality and Missing Values

Column	Missing entries	Reported handling
Age	177	Imputed
Cabin	687	Retained as missing due to high sparsity
Embarked	2	Imputed

Cabin has 687 missing entries—approximately 77.1% of the dataset—so it provides limited coverage without additional processing. Age and Embarked were reported as imputed; the exact imputation method was not specified in the supplied findings.

4. Age Distribution

- Passenger ages range from approximately 0.42 years (about five months) to 80 years.
- The reported mean age is approximately 29.36 years, while the median is 28 years.
- The distribution is described as slightly right-skewed, with many passengers between 22 and 35 years.
- A smaller peak among young children aged 0–5 years is also reported.

Interpretation: Most passengers were young adults, but the dataset also included infants, children, and elderly passengers. A smaller number of older passengers may pull the mean above the median.

5. Survival Analysis

Overall survival

Outcome	Passengers	Share of dataset
Survived	342	38.38%
Did not survive	549	61.62%

Survival rate by passenger class

Passenger class	Reported survival rate
1st class	62.96% — highest
2nd class	47.28%
3rd class	24.24% — lowest

Interpretation: Survival rates were higher among first-class passengers and lower among third-class passengers. This is an association in the observed data; class alone does not establish the cause of an individual's outcome.

Survival rate by gender

Gender	Passengers	Reported survival rate
Female	314	74.20%
Male	577	18.89%

Interpretation: Female passengers had a substantially higher observed survival rate than male passengers in this dataset.

6. Relationships Between Variables

Age and fare

The reported scatter plot did not show a strong direct relationship between age and ticket fare. The analysis notes that higher-fare passengers tended to show higher survival across age brackets. This is a visual pattern, not proof that fare directly caused survival.

Reported numerical correlations

Variable pair	Correlation	Interpretation
Fare & Survived	+0.26	Positive association
Pclass & Survived	–0.34	Higher class number associated with lower survival
Pclass & Fare	–0.55	Higher class number associated with lower fare
SibSp & Parch	+0.41	Positive association between family-related counts

Note: Pclass is coded from 1 (first class) to 3 (third class), so a negative correlation with Survived means larger class codes are associated with lower survival. Correlation does not, by itself, demonstrate causation.

7. Outliers and Limitations

- Fare: The maximum reported fare is 512.33, well above the reported 75th percentile of 31.00. This may represent a small number of unusually high fares.
- Age: Passengers aged 70–80 were identified as potential upper-end outliers. These are valid observations and should not automatically be removed.
- Cabin: About 77.1% of Cabin entries are missing, limiting cabin-specific analysis.

- Scope: Findings are descriptive and may be affected by missing data or the chosen imputation method. They do not prove causation.

8. Conclusion

The EDA shows that 38.38% of the 891 passengers in the dataset survived. Survival was substantially higher among female passengers and first-class passengers, while third-class passengers had the lowest reported survival rate. The dataset contains incomplete Age, Cabin, and Embarked information; in particular, the high proportion of missing Cabin values limits its usefulness. Reported correlations indicate relationships among fare, passenger class, and survival, although correlation should not be interpreted as causation. Overall, the analysis demonstrates how data summaries and visual exploration can reveal important patterns and data-quality issues before further analysis.

Report note: Values and findings are based on the summary provided for this project. Verify numerical results against the final notebook if you make further changes to the analysis.